

Task 3: Protein Structure Visualization Report

The three-dimensional (3D) structure of the selected protein was visualized using UCSF Chimera, a molecular modeling and visualization tool widely used in bioinformatics. The protein structure was obtained in PDB format from the AlphaFold Protein Structure Database. Upon loading the structure into Chimera, the ribbon (cartoon) representation was applied to better visualize the secondary structural elements of the protein. The visualization revealed that the protein exists as a homodimer, consisting of two identical alpha chains. Each chain is predominantly composed of alpha-helical structures, which are visible as spiral ribbons in the model. These alpha helices indicate a stable, compact, and globular protein conformation. Different colors were used to distinguish between the two chains, allowing clear identification of their spatial arrangement and interaction. This helps in understanding how subunits assemble to form the functional protein. Overall, visualization of the protein structure provides valuable insights into its folding, structural organization, and potential functional properties. Tools like UCSF Chimera play an important role in analyzing biomolecular structures in modern bioinformatics.